Exam Date & Time: 27-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

INTRODUCTION TO AIRCRAFT STRUCTURES

INTRODUCTION TO AIRCRAFT STRUCTURES [AAE 2121]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

Answer all questions.
Assume any missing data with proper reasoning.

1A) How would you apply bending stress analysis to evaluate the structural integrity of an aircraft wing under aerodynamic loads? (2)

1B) Prove that the bending stress in any layer is proportional to the distance of that layer from neutral axis in a beam. (4)

1C) A cast iron T section beam shown in figure 1 is simply supported on a span of 8 m. The beam carries a uniformly distributed load of 1.5 kN/m length on the entire span. Evaluate the maximum tensile and compressive stresses. (4)

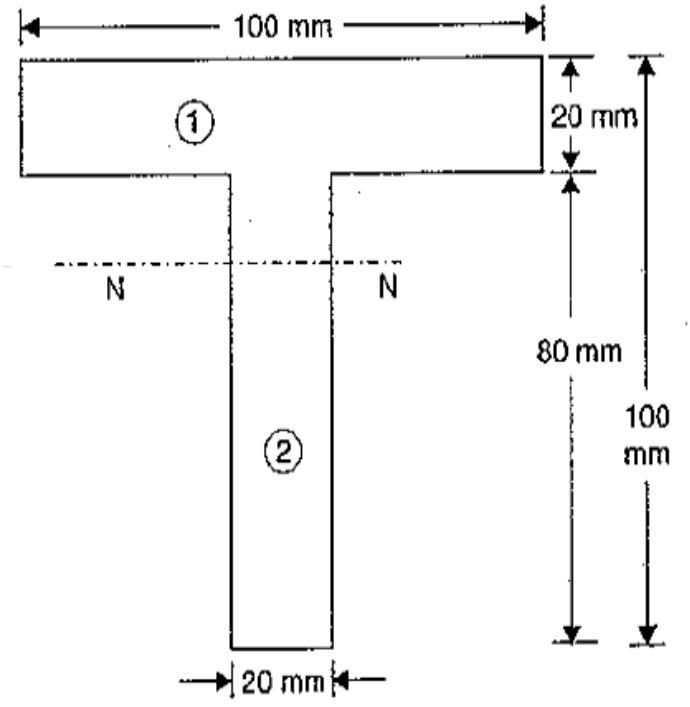


Figure 1.

2A) Analyse the conditions under which the radius of the Mohr circle becomes zero. What does this signify? (2)

2B) Derive an expression for the stresses on an oblique plane of a rectangular body, when the body is subjected to a simple shear stress. (4)

2C) A point in a strained material is subjected to the stresses as shown in figure 2. Locate the principal planes and evaluate the principal stresses. (4)

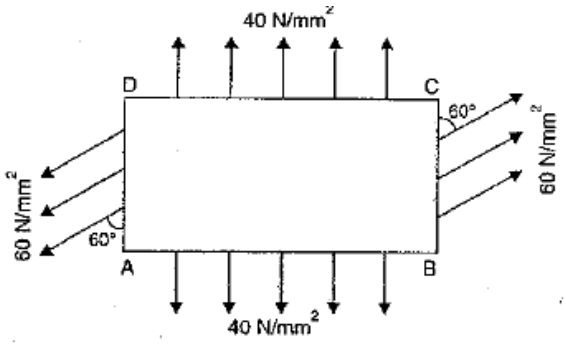


Figure 2.

3A) Explain the significance of analysing beam deflection and slope in aircraft structural components. (2)

3B) A beam ABC of length 9 m has one support at the left end and the other support at a distance of 6 m from the left end. The beam carries a point load of 12 kN at right end and also carries a udl of 4k N/m over a length of 3 m as shown in figure 3. Determine the slope and deflection at point C take E= 2×10^5 N/mm^2 and I = 5×10^8 mm^4. (4)

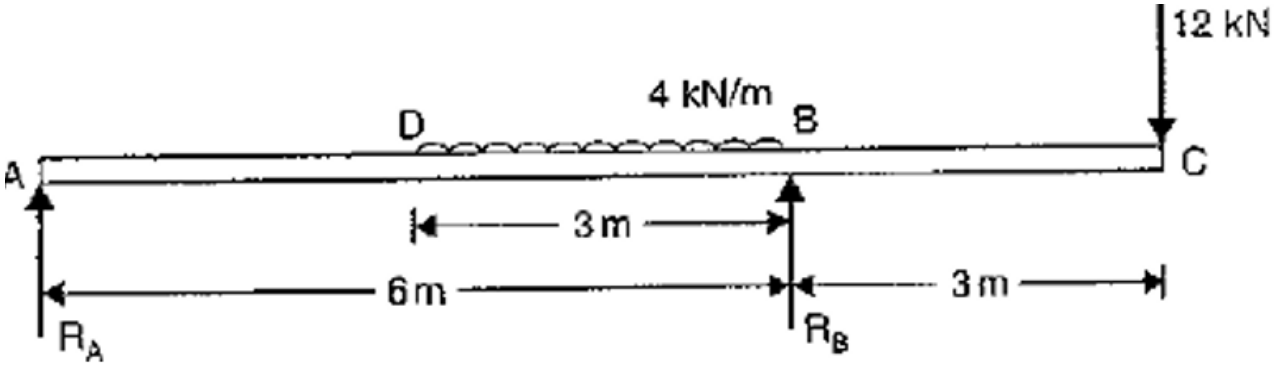


Figure 3.

3C) Evaluate and draw shear force, bending moment diagrams for the single side overhanging beam subjected to loading as shown in figure 4. Also, mark salient points on SFD and BMD. (4)

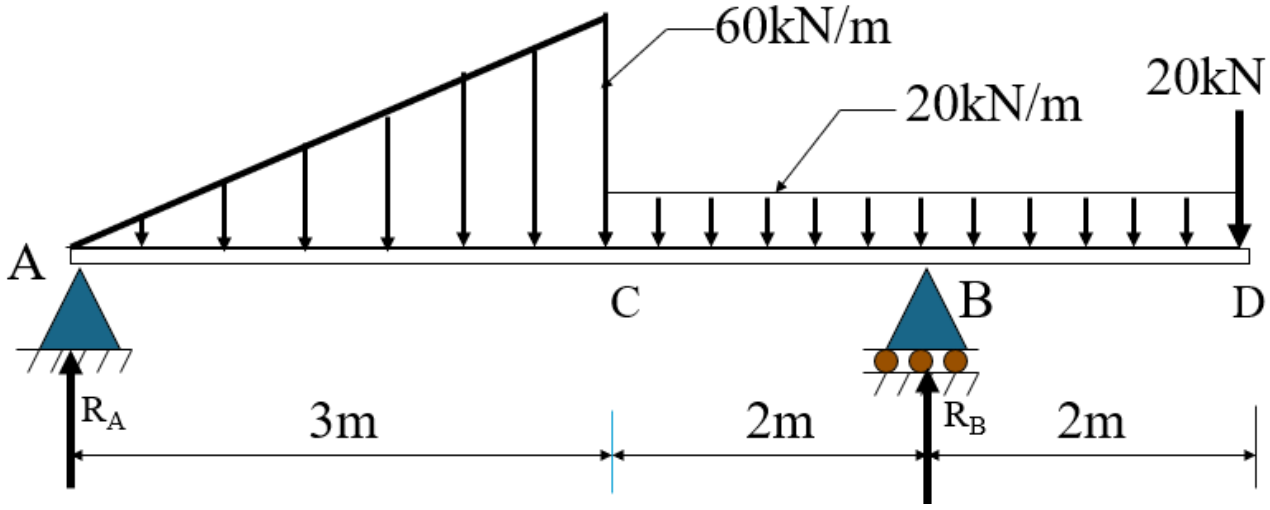


Figure 4.

4A)	Define torsional stress and explain how it is generated in a solid shaft.	(2)
4B)	Two shafts of the same material and of same lengths are subjected to the same torque, if the first shaft is of a solid circular section and the second shaft is of hollow circular section, whose internal diameter is 2/3 of the outside diameter and the maximum shear stress developed in each shaft is the same, compare the weights of the shafts.	(4)
4C)	Determine the changes in length, breadth and thickness of a steel bar which is 4 m long, 30 mm wide and 20 mm thick and is subjected to an axial pull of 30 kN in the direction of its length. Take E=200 GPa and Poisson's ratio of 0.3.	(4)
5A)	Why do engineers use the concept of equivalent length when analysing the buckling behaviour of columns?	(2)
5B)	Explain the need for failure theories. Provide a brief overview of the maximum strain energy theory and the maximum shear strain energy theory.	(4)
5C)	Derive an expression for the buckling load of a column with one end fixed and another end free.	(4)

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